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REVIEW ARTICLE

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Enhancing monitoring of mangrove spatiotemporal tree diversity and distribution patterns

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Abstract

Spatiotemporal information on mangrove species assemblage of natural, disturbed, and rehabilitated is an essential prerequisite for effective strategies for biodiversity conservation and management. However, appropriate field-based sampling strategies of spatial heterogeneity still hamper the detection of the species distribution and its temporal development. An increasing amount of remote sensing data seems the perfect way to tackle these challenges. With this article, we fill this gap by presenting a review of the challenges and limitations to assess the current status of species diversity. We conclude that species discrimination based on remote sensing techniques is still limited by atmospheric contamination and tidal fluctuations. The lack of accurate information on the spatiotemporal development of species diversity and forest structure further curtails an understanding of functional indicators and the predictive power of modeling approaches. Nevertheless, multi-source remote-sensing techniques could seemingly capture the landscape heterogeneity and support systematic sampling designs. Spatially balanced (systematic) training and validation data are necessary to compile robust spatiotemporal information, supporting reliable predictions for optimizing restoration efforts. A systematic sampling of spatiotemporal ecological information is vital to derive the historical state of mangroves, detecting their degradation, and predicting future patterns of species distribution that are generally crucial for restoration, and particularly to rehabilitate species diversity.

KEYWORDS
diversity prediction, field inventory, mangrove species diversity, systematic sampling, tidal influence, trajectories

1 | INTRODUCTION

Mangrove forests occupy only 0.5% of the world's coastal areas, between 32° N to 38° S (Giri et al., 2011; Saenger et al., 2019). Mangrove forests comprise a dominant group of woody trees (~70 species), shrubs, and fern species across 25 genera (19 families) that are distributed in the coastal intertidal areas of global tropical and subtropical regions (Duke, 2017; Tomlinson, 2016). A pronounced gradient of tree species diversity from the new world (17 species) to the Indo-West-Pacific is observed (IWP; 54 species; Duke, 2017; Ellison et al., 2020; Tomlinson, 2016). Mangrove diversity also greatly varies at different spatial scales, that is, regional, estuarine, and intertidal (Saenger et al., 2019), since tree species occurrence is strongly linked to ecological control factors (Tomlinson, 2016; Twilley et al., 2017).

The extent of mangroves has decreased by 1.04 million hectares between 1990 and (FAO, 2020), and ~16% of mangrove species are at an elevated threat of extinction (Polidoro et al., 2010). Although mangrove deforestation rates appear to be in decline, countries like

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the Philippines, Cambodia, Indonesia, Myanmar, and Malaysia are still facing high losses (Friess et al., 2019; Gandhi & Jones, 2019; Goldberg et al., 2020). Between 2000 to 2016, 62% of global mangrove losses, and 80% of the losses within Southeast Asia, were the direct result of the replacement with aquacultural and agricultural enterprises (Goldberg et al., 2020). Notably, these areal losses in mangrove coverage are accompanied by a decline in species diversity (Macintosh & Ashton, 2002). Moreover, the majority of rehabilitation projects in Southeast Asia implement monoculture plantations (usually, *Rhizophora apiculata*; Ellison, 2000). The conversion of mangrove forests to shrimp farms or agriculture alters the topography or natural elevation gradients, thereby changing the controlling biophysical variables of species distributions (Hickey & Bruce, 2010). Among other stressors, shrimp farm waste or effluent from industries can detrimentally impact ecological function and species diversity (Capdeville et al., 2018; Vaiphasa et al., 2007).

Mangroves are now considered a high-priority ecosystem for international conservation efforts, such as the Blue Carbon Initiative and the Global Mangrove Alliance (Friess et al., 2020). Considering the historical exploitation and replacement with monoculture plantations, a loss of species would pose severe ecological and economic consequences, particularly for regions with low species diversity and elevated rates of species loss. To date, the inclusion of the factors mentioned above has rarely been considered when assessing the spatial distribution of mangrove and their species diversity.

Early documentation reports patterns perpendicular to the coastline related to changes in the frequency of tidal inundation on the West Malaysia (Watson, 1928). Nevertheless, zonation patterns are somewhat conceptual, and arbitrary inundation classes have been proposed (Ellison, 2002; Tomlinson, 2016). Moreover, site-specific inundation categories may not properly fit species distribution at larger scale, particularly where intertidal and spatial patterns are intrinsically complex (Buckley, 1982; Bunt & Williams, 1980; Peters et al., 2020; Pimple et al., 2022). The strong zonation patterns are rarely observed in the field and larger scale distribution patterns remain poorly understood (Bullock et al., 2017; Ellison et al., 2000). Ellison (2002) identified four possible explanations why many studies have failed to observe spatial distribution patterns: (1) lack of proper statistical methods for testing; (2) lack of appropriate spatial scales for observation and analysis, (3) failure to consider human impacts; and (4) zonation may not actually exist in mangrove forests (the 'null hypothesis').

The tree density, species composition, and structural diversity vary significantly in natural, naturally regenerated, and rehabilitated mangrove forests (Asaeda et al., 2016; Azman et al., 2021; Pimple et al., 2022). For example, most rehabilitation projects implement monoculture plantations consisting of *Rhizophora* spp. since this species can be easily nursed and handled. As a result, the rehabilitated stands are less complex in diversity and structure than natural forests. On top, such stands often fail in the mid or long-term in cases where *Rhizophora* spp. is less adapted to the particular environmental settings (Kodikara et al., 2017) than other mangrove species which would occur naturally at the given location. In contrast, the species diversity and structural complexity of naturally regenerated stands may vary based on the seed availability and the particular environmental settings at the given location (Kairo et al., 2002; Pimple et al., 2022). For example, limited tidal inundations may increase pore water salinity favoring species that are more salt tolerant than *Rhizophora* spp. To date, the degree of success of rehabilitation or regeneration efforts is often assessed in terms of the number of planted seedlings or in the size of the regenerated area. However, the long-term changes in diversity patterns are rarely considered or reported during diversity assessments across different spatial scales, as most quantitative baseline studies use single surveys at a single point in time while neglecting temporal changes in species diversity (Abino et al., 2014; Joshi & Ghose, 2014; Nehru & Balasubramanian, 2018; Shah et al., 2016). Lastly, a few mangrove ecosystems worldwide are uninfluenced by human activity (Ellison, 2002). Thus, it remains uncertain whether rehabilitated, restored, and regenerated mangroves have truly recaptured the preexisting species diversity before degradation. In this sense, simply assessment of total loss rates or changes in the areal extent of mangrove forests is inadequate for conservation efforts without sufficient information on the spatial distribution of the remaining habitat (Bryan-Brown et al., 2020). If the purpose is to conserve or restore natural mangrove forests in their full functional diversity and with the original ecosystem services, it is thus important to monitor the tree species diversity, and the structural complexity of the naturally regenerated and rehabilitated stands and to compare these features with the natural forests. Moreover, species-specific information, as well as information on spatiotemporal changes in mangrove species diversities, is urgently needed to refine conservation practices for threatened mangrove species (Polidoro et al., 2010).

Remote sensing provides a unique opportunity to characterize, map, and monitor spatiotemporal changes across the landscape, at scales ranging from individual trees to groups of dominated species or spatial characteristics (Cárdenas et al., 2017; Heumann, 2011; Kuenzer et al., 2011; Lucas et al., 2017; Pham et al., 2019; Wang et al., 2019). Advances in remote sensing technologies show promising opportunities for designing systematic sampling and mapping the spatiotemporal distribution of mangrove species. Wang et al. (2019) identified key milestones in applying remote sensing techniques for mangrove species mapping, noting that the identification of species or their associations at different landscape scales was not yet fully defined. This is due to, depending on the site-specific conditions, mangrove tree species can take one of the following forms: (i) single species zone, where a single tree species monoculture zone (natural or rehabilitated) is systematically arrayed over the landscape; (ii) grouped zone, where ≥2 species are grouped across the landscape; (iii) randomly distributed or no zone, where species are randomly placed over the landscape (Ellison et al., 2000; Pimple et al., 2022; Rivera-Monroy et al., 2017). Nevertheless, species-level discrimination using remote sensing requires a detailed field information about habitat conditions (Kuenzer et al., 2011) and quantification of diversity characteristics at the landscape scale. In addition, optical (spectral) and synthetic aperture radar (SAR) characterization and discrimination of mangrove species can be influenced by various atmospheric

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contaminant and tidal fluctuations (Kuenzer et al., 2011; Rogers et al., 2017). Furthermore, an understanding of temporal changes in rehabilitated and regenerated mangroves is crucial. In a single date information, spectral confusion among species or their associations may arise across different forest stands. The integration of consistent time-series could be an efficient way to characterize spatially explicit information on forest attributes such as rehabilitated, naturally regenerated, and natural forest stands. With the comprehensive sampling, the radiometrically stable spectral data and temporal information will aid to improve the mapping of complex species diversity patterns. Spatial explicit information on species diversity can be very useful criteria to evaluate diversity baseline under different management practices.

To date, many remote sensing and field studies have failed to report reliable diversity distribution patterns due to a lack of systematic field data collection, quantification and prediction of species diversity patterns in relation to various environmental gradients, and assessments regarding the effects of conservation and other anthropogenic interventions on successional diversity. The subsequent problems also apply to the usage of recent earth observation data for mapping and monitoring mangrove forest species diversity as long as they do not overcome the effect of various influences and integrate species-specific spectral characterization. This study aims to: (1) identify what are the current challenges and importance of systematic field data for diversity quantification; (2) understand which key factors are affecting remote sensing data quality and how to overcome them to study mangrove species diversity; (3) discover what are the key limitations and the potential of the remote sensing in predicting (modeling) species diversity; (4) highlight what are the key processing steps that could enhance the remote sensing-based diversity mapping and monitoring.

2 | MATERIALS AND METHODS

This article is a literature review supported by remote sensing observation using reproducible examples from a site in eastern Thailand (https://rescue-pro.net/). We used keywords related to the mangrove forest species diversity studies and expressions that refer to ecological field studies, remote sensing, and modeling using the search engines of Scopus, Google Scholar, and Web of Science. The keyword combinations that were used included ("mangrove" AND "species" AND "diversity" AND "zonation") AND ("rehabilitation" OR "plantation") AND/OR ("remote-sensing" AND "time-series" OR "trajectory") AND ("systematic" AND "stratified" AND "random" AND "sampling") AND ("mangroves" AND "individual-based" AND "modeling" OR "prediction" AND "remote-sensing"). We also included separate searches that combined "remote-sensing" with a specific focus on the effect of environmental (e.g., tidal inundation) and atmospheric contaminants (e.g., cloud cover) on multispectral and SAR data. Peer-reviewed journal articles published in English were examined. We selected for inclusion in the review article only those which explicitly identify the challenges associated with limitations of (1) current field inventories, (2) optical (spectral) and SAR (backscatter) imagery, (3) time series data, and (4) predicting species diversity. Further to overcome current limitations, we reviewed journal articles from 2012–2022 that explicitly identify the importance of (1) systematic sampling and required environmental data, (2) error-free and consistent quality of remote sensing data, (3) time series data for temporal change and stages of succession, and (4) spatiotemporal dynamics of forest diversity and structure for enhancing diversity prediction studies. Many research articles in this period reflect the growing need for using good practices for using field and remotely sensed data. Finally, based on our literature review and expertise in the field of mangrove ecology, biodiversity concept, remote sensing, and modeling, we developed a checklist desirable dataset that will be useful in enhancing the monitoring of mangrove spatiotemporal tree diversity.

3 | CHALLENGES TO SPECIES DIVERSITY MONITORING AND PREDICTIONS

3.1 | Field inventory data: as the base for remote sensing approaches

So far, studies focusing on mangrove species diversity are often limited to field inventories, reporting species frequencies or abundances measured along transects and extrapolating it to landscape-scale. In most cases, species occurrence cannot be directly related to environmental factors (Ellison, 2002; Ma et al., 2020). The tidal fluctuations, soft muddy substratum, and complex forest structure are posing significant challenges for logistics as well as hindering the measurement of these factors at the locations of single trees. Instead, soil factors are usually provided as averaged values over forest landscapes, although topography could serve as a proxy for salinity, tidal inundation, and edaphic characteristics, among other factors (Hickey & Bruce, 2010). The inventory surveys are conducted either on fixed areas, randomly chosen quadrat plots, or along transects perpendicular to the coastline, for instance, using the point center quarter method. In the majority of the examined studies, information about the criteria to choose the location of transects and plots and accountability of spatial species heterogeneity are not stated (but see Ellison et al., 2000; Pimple et al., 2022); thus, such a sampling approach could be considered unsystematic. Stratified or systematic sampling designs, as known from other forest inventory sciences (Köhl et al., 2006; Olofsson et al., 2014; Wallner et al., 2018), are usually not explicitly mentioned so that prior planning of a survey considering spatial heterogeneity and vegetation patterns prominently shown as mangrove zonation cannot be assumed. Further, reporting species distribution in relation to environmental gradients (Table 1) and historical status of diversity change is not a common practice. This is crucial to understand the species assemblage in a given site. Although a complete user-defined stratified random sampling approach conversing every small area within the forest can undoubtedly account for spatial heterogeneity (Ellison et al., 2000), it requires a huge sampling effort, providing little, if any, information on rehabilitation and regenerated

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TABLE 1 Species diversity studies conducted at the high-diversity regions that were reviewed in the frame of this study

Sampling method	Environmental factors	Study site	References
Transect line (random plots)	-	New Hebrides Island	Marshal & Medway (1976)
Transect line (random plots)	-	Cape York Peninsula, Australia	Bunt (1996)
Transect line (random plots)	-	Sarawak, Malaysia	Shah et al. (2016)
Transect line (random plots)	Elevation	Zhenzhu Bay, Guangxi, China	Ma et al. (2020)
Transect line (random plots)	-	Mumbai, west coast, India	Kantharajan et al. (2018)
Transect line (random plots)	-	Andaman and Nicobar Islands, India	Ragavan et al. (2015)
Transect line (random plots)	-	Bay Islands, Honduras	Fickert & Grüninger (2010)
Transect line (equal intervals and arbitrary zones)	-	Northern Queensland	Bunt & Williams (1980)
Fixed area transect line	Elevation	Ngomeni, Mombasa and Gazi, Keneya	Ruwa (1993)
Fixed area transect line	N, P, K, pH, and salinity	Gorontalo, Indonesia	Baderan et al. (2018)
Fixed area transect line	-	Klang Islands, Selangor, Malaysia	Rozainah et al. (2018)
Fixed area transect line	-	Southwest coast of India	Sreelekshmi et al. (2018)
Fixed area transect line	pH, TC, TN, DO, Eh, Sand, Soil temperature, silt	Sundarbans, India	Sreelekshmi et al. (2020)
Stratified random fixed area transect line	Elevation, NaCl, CEC, FC, silt	Sundarbans, Bangladesh	Ellison et al. (2000)
Stratified random sampling (manual selection)	-	Yaedashae, Gyobingauk, Nattalin, Pauk, Seikphyu, Myanmar	Khaine et al. (2017)
Quadrat sampling	-	Botoc, Samar, Philippines	Abino et al. (2014)
Quadrat sampling	-	Segara Anakan, Cilacap, Central Java, Indonesia	Widyastuti et al. (2018)
Belt transect plot	-	Nicobar Islands, India	Nehru & Balasubramanian (2018)
Point-centred quarter method	-	Kakinada Bay, Andhra Pradesh, India	Satyanarayana et al. (2002)

Note: The listed environmental factors are given as average values at the plot level. The temporal dynamics were not monitored in any of these studies. Abbreviations: CEC, cation exchange capacity; DO, dissolved oxygen; Eh, redox potential; FC, field capacity; SM, soil moisture content; TC, total carbon; TN, total nitrogen.

forest diversity. Remote sensing-based approaches and earth observation imagery can support the identification of the field survey plots by providing long-term information on the status of the forest as well as spatial scale species heterogeneity. Thus, such techniques can complement field surveys and thus improve the large spatial scale assessment of mangrove species distribution and related ecological features (Pimple et al., 2022) while minimizing sampling efforts and associated costs.

3.2 | Remote sensing: Analysis of species occurrence simultaneously with environmental factors at multiple points in time and various spatial scales

Remote sensing has proven useful for characterizing the extent of mangrove habitats, their spatial distribution (discrimination of mangrove species or communities), and temporal changes (gain and loss) resulting from various stressors or conservation efforts. Moreover, there are several challenges associated when mapping over larger spatial scales. For example, the spectral reflectance and backscatter properties are influenced by tidal inundation. Furthermore, despite significant advancement in capacity, temporal dynamics of species diversity has rarely been considered.

3.2.1 | Challenges of spectral reflectance in multispectral imagery

Frequent dense cloud cover and cloud shadow over coastal and mangrove areas are one of the first issues encountered when selecting optical satellite imagery. Cloud cover, shadows, haze, and missing values can influence several analysis processes, leading to the misidentification of species and their temporal dynamics. In cloudy coastal regions, multispectral sensors might only provide a few, or perhaps no cloud-free images throughout the year, thereby limiting the monitoring potential of mangrove ecosystems (Figure 1). Some researchers have only performed cloud pixel masking, leaving the shadowed pixels untreated (Islam et al., 2019), yet cloud shadows change the spectral quality of affected pixels. Accordingly, the challenges of cloud cover and shadows over mangrove habitats have been well-documented in

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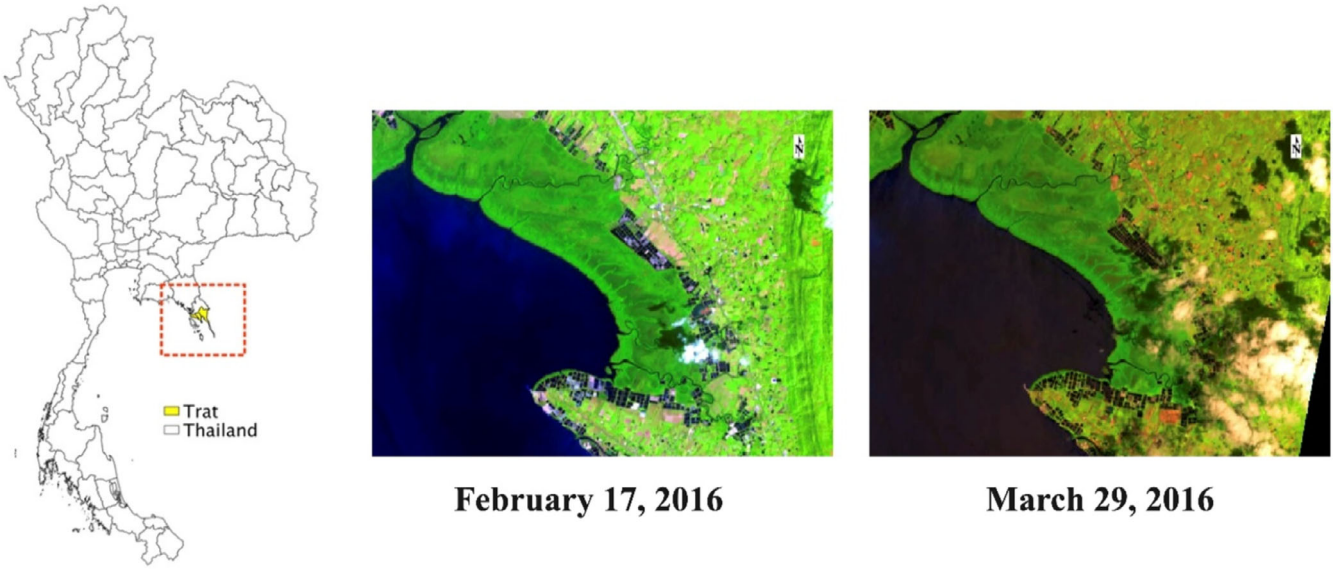

FIGURE 1 Example of a cloudy region throughout the year. The illustration of Landsat 8 best available single date cloud-free image in 2016 over Trat Province, Thailand in pseudo-colour-composite (SWRI-1, NIR, RED). The available clouds and shadows limit the consistency of available images required for continuous change monitoring [Colour figure can be viewed at wileyonlinelibrary.com]

previous studies (Cárdenas et al., 2017; Kuenzer et al., 2011). Despite these challenges, numerous methods to automatically screen clouds and cloud shadows from optical images have been developed recently with the support of cloud computing platforms such as Google Earth Engine (Pimple et al., 2020; Pimple et al., 2022; Simonetti et al., 2021; Zhu & Helmer, 2018).

Another challenge in mapping species level diversity at varying spatial scales is the periodic presence of water beneath the mangrove canopy (Kuenzer et al., 2011; Lucas et al., 2017; Rogers et al., 2017). The spectral reflectance of open canopy mangroves (such as submerged, sparse, and scrub) is more strongly altered by high and low tidal inundations compared to closed canopy mangroves (Rogers et al., 2017; Xia et al., 2018). This phenomenon further complicates the detection of mangrove distribution patterns across different temporal scales, particularly if adjacent cloud-free scenes have been acquired at different tidal stages (Rogers et al., 2017). Further, the effects of tidal levels on spectral reflectance values in mangrove forests are poorly quantified. Jusoff (2006) investigated the sensitivity of the NIR range to mangrove species discrimination. Depending on the amount of crown cover and level of inundated water (beneath canopy), the altered spectral reflectance can also be observed in closed canopy forests due to the strong NIR absorption of water, weakening the performance of multispectral imagery. Studies that analyzed multi-tide and multi-density (height and canopy cover) imagery of different dates revealed the spectral overlapping and reduced NIR values in open and closed canopy stands (Perea-Ardila et al., 2021; Zhang & Tian, 2013).

The benefits of high-resolution imagery for mangrove species identification have been recognized (Lucas et al., 2017), although the influence of high and low tidal inundations have been neglected (Heenkenda et al., 2014; Huang et al., 2009; Neukermans et al., 2008; Rahman et al., 2019; Wang et al., 2016; Wang et al., 2018). Furthermore, most studies attempting mangrove species identification via medium-resolution satellite imagery data neither account for the tidal influence (Bullock et al., 2017; Ghosh et al., 2016; Nayak & Bahuguna, 2001; Valderrama-Landeros et al., 2018; Wang et al., 2018).

Recent studies have tested the performance of high- and medium-resolution multispectral imagery in both low and high tide conditions, identifying several potential improvements (Cárdenas et al., 2017; Proisy et al., 2018; Rogers et al., 2017; Xia et al., 2018; Zhang et al., 2017). Rogers et al. (2017) reported that spectral properties could become increasingly homogeneous (similar spectral response from two different species) under specific tidal conditions. Further, if the two single date images were acquired in different tidal conditions, heterogeneous spectral responses may be observed for the same species. Xia et al. (2018) also found that the spectral signatures of seaward low-stand mangroves varied greatly during high and low tide. Furthermore, multi-date analyses combining high and low tide imagery may result in the under- or overestimation of mangrove extent or species areas (Zhang et al., 2017). Proisy et al. (2018) and Cárdenas et al. (2017) noted that the image classification accuracy and consistency depend on environmental conditions such as tide level. The above review suggests that varied spatiotemporal data aids in species discrimination and mapping accuracy; however, these are highly dependent upon image acquisition time and tides.

Figure 2 presents examples of the effects of high and low tides on imagery observed in the Trat Province, Thailand. The seaward location (orange circle) and the mid-zone (black circle) were dominated by *Sonneratia abla–Rhizophora* sp. and dense scrub *Ceriops tagal*, respectively. Panels (a–b, e–f) show the observed locations and reflectance values of LANDSAT-8 (OLI) during high and low tide, revealing that

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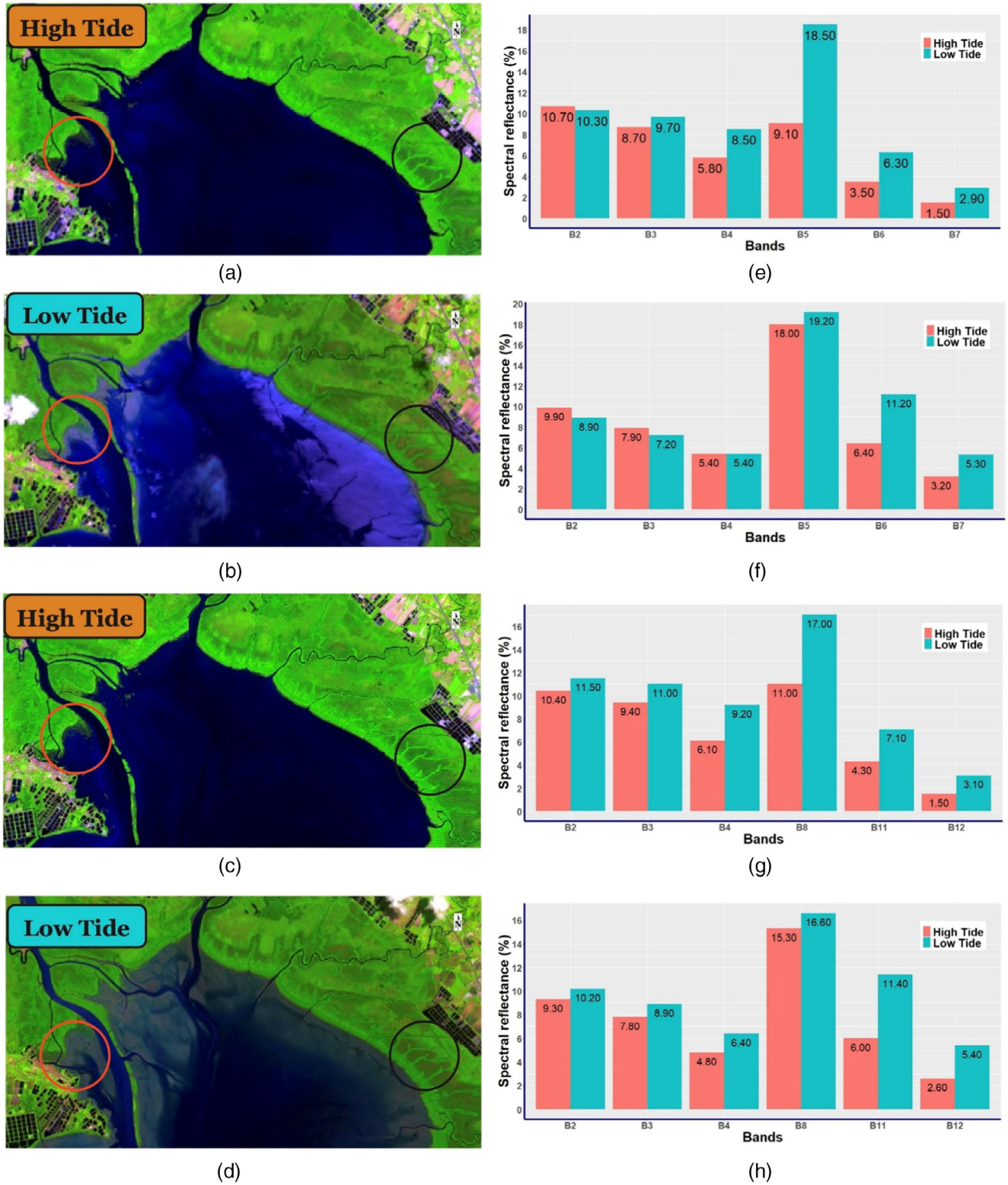

FIGURE 2 Effects of tidal inundation on the reflectance of: (a,b) LANDSAT-8 (OLI), and (c,d) SENTINEL-2 (MSI) TOA imagery. Satellite acquisition time was ~10:00–10:30 (local hour), Trat Province, Thailand, and present the: (a,c) present false-colour composite (SWIR-1, NIR, and red) of LANDSAT-8 (OLI) and SENTINEl-2 (MSI) images (January 27, 2017, and December 25, 2018, respectively) at high tide (3.49 m and 3.36 m); whereas (b,d) show similar false-colour composites images (may 10, 2017, and may 4, 2018, respectively) at low tide (2.00 and 1.76 m). Further, (e, f) display the spectral variability of LANDSAT-8 imagery at seaward and scrub stand (i.e., mid-zone; orange and black circle, respectively) locations during tidal fluctuations; where (g,h) present the same spectral variability for SENTINEL-2. Notably, the graphed spectral reflectance values are scaled by 100 for visualization [Colour figure can be viewed at wileyonlinelibrary.com]

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during low tide, the spectral reflectance values of NIR, SWIR-1, and 2 bands were higher than at high tide. Alternatively, panels (c–d, g–h) show the observed locations and reflectance values of SENTINEL-2 (MSI) during high and low tide. Similarly, the low tide NIR, SWIR-1, and 2 bands values of SENTINEI-2 were higher. These results indicate that a consistent spectral signature is crucial for spatiotemporal mangrove species research to avoid the spectral interference related to the tidal presence of water. Thus, images should be acquired at the same tidal level (i.e., low tide) whenever possible to preserve the spectral properties of species or their associations at given site.

3.2.2 | Challenges of backscatter in synthetic aperture radar (SAR) imagery

Microwave radar, in particular SAR at the X, C, L, and P bands have been used to identify different mangroves (Held et al., 2003; Mougin et al., 1999; Simard et al., 2002). Unlike multispectral sensors, SAR sensors are capable of forest landscape observations by day or night, regardless of cloud cover or nearly any other type of weather condition (Ottinger & Kuenzer, 2020). However, SAR backscatter in mangrove forests is sensitive to below canopy inundation or open water in general (Sahour et al., 2022). Wang and Imhoff (1993) modelled L-band radar backscatter from mangrove forest stands, reporting higher values in inundated, compared to non-inundated conditions. Darmawan et al. (2015) investigated the impact of tidal inundation on ALOS PALSAR polarimetric measurements of HH horizontal transmit and horizontal receive (HH), and horizontal transmit and vertical receive (HV), similarly revealing the effect of the tidal regime on the backscatter coefficient. Lucas et al. (2017) noted the effects of tidal inundation on L-band HH polarization, as water under the canopy leads to a strong double-bounce interaction between the water surface and tree trunks.

Partially or fully submerged mangroves and open canopy vegetation areas, such as sparse and scrub forest stands on the seaward and low elevation areas (inundated during high tide), are likely to show such variations in satellite observations (Figure 3). Lucas et al. (2017) also found that backscatter was higher with increasing aboveground biomass (up to a frequency-dependent saturation point), further indicating the sensitivity of SAR observations over inundated open canopy forest. Further, several studies have recognized SAR data as complementary to optical imagery for mangrove assessment (Lucas et al., 2020; Simard, 2019; Zhang et al., 2018).

Herein, the literature pertaining to the application of SAR and combined multispectral imagery with SAR for mangrove species identification was compiled and reviewed, including WORLDVIEW-3 and RADARSAT-2 (Zhang et al., 2018); ALOS PALSAR, ALOS PRISM, and IKONOS (Brown et al., 2016); ALOS PALSAR-2 and RADARSAT-2 (Ferrentino et al., 2020); WORLDVIEW-2 and TANDEM-X SAR (Rahman et al., 2019); and SENTINEL-1 SAR and SENTINEL-2 MSI (Xia, Yokoya, & Pham, 2020). Notably, none of these studies considered the tidal influence on SAR backscatter values.

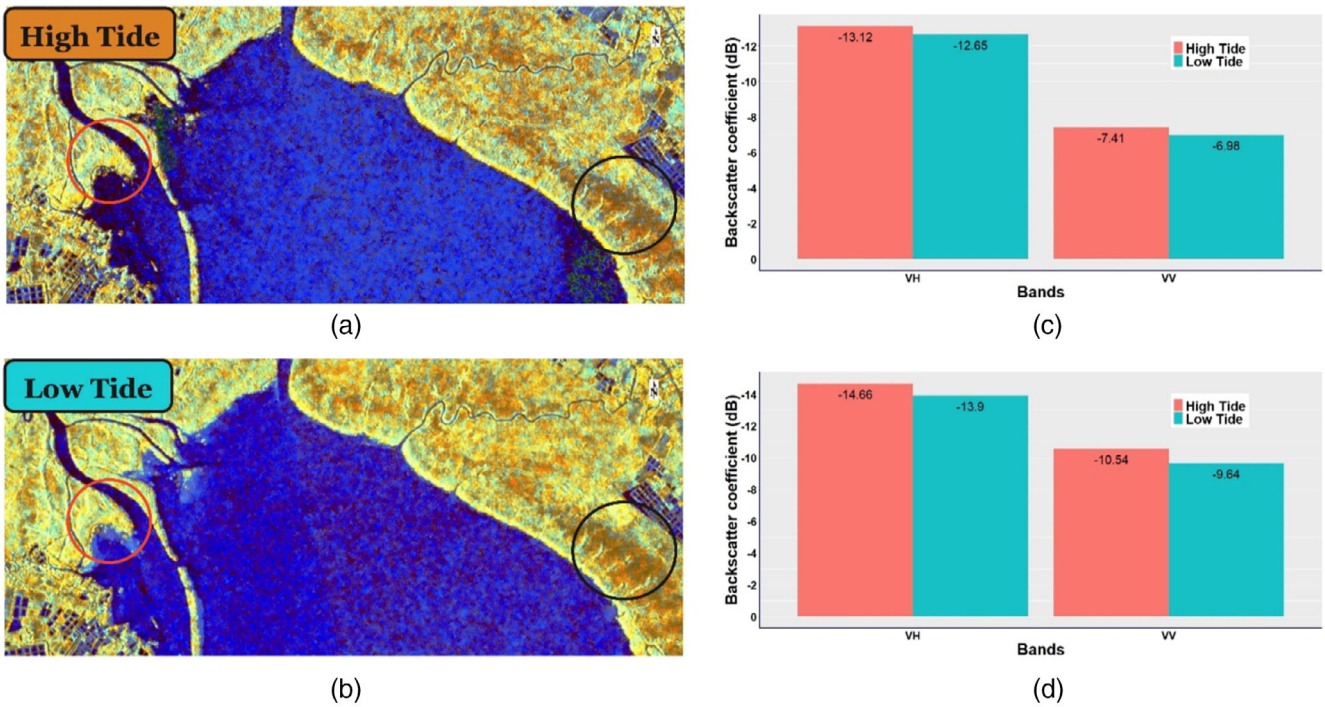

FIGURE 3 Effects of tidal inundation on the backscatter coefficient of SENTINEL-1 SAR imagery. Satellite acquisition time was ~11:00 (local hour), Trat Province, Thailand. Maps (a,b) display RGB composites (VV, VH, and VV/VH) of SENTINEL-1 SAR images (December 26, 2018; June 17, 2018, respectively) at high and low tide (3.35 and 1.12 m, respectively). Graphs (c,d) portray the backscatter coefficient variability at seaward and mid-zone locations (orange and black circles, respectively) during tidal fluctuations [Colour figure can be viewed at wileyonlinelibrary.com]

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3.2.3 | Temporal inconsistencies in mangrove assessment

The inconsistencies in current mangrove databases contributed to incomplete baseline reference scenarios (Hamilton & Casey, 2016). Despite the importance of multi-temporal changes in mangrove forest management and tree species distribution monitoring, required optical sensors data remain limited due to the differences in local environmental (tidal inundation)and atmospheric conditions (clouds and their shadows), in addition to the high variability of mangrove ecosystems. SAR data can be used regardless of various atmospheric contaminants, although this data lacks the temporal frequencies and ability to differentiate species (Kuenzer et al., 2011; Lucas et al., 2017).

The number of satellite sensors and the platforms for big data storage and processing have increased over the past decades (Gomes et al., 2020; Kuenzer et al., 2014) including (1) LANDSAT-8 OLI, LANDSAT-9 OLI-2, SENTINEL-1 SAR, −2 MSI, and PLANETSCOPE, are capable of producing larger scale imagery with high temporal frequencies (Fassnacht et al., 2016; Francini et al., 2020; Masek et al., 2020), (2) upcoming SAR satellites, such as NISAR (L- and S-band sensors), a joint National Aeronautics and Space Administration (NASA) and Indian Space Research Organization mission scheduled to launch in late 2022, as well as the European Space Agency's BIOMASS (P-band sensor), also scheduled to launch in late 2022, can increase the capability of characterizing and monitoring mangrove ecosystems. Finally, ongoing developments in satellites and airborne sensors, including light detection and ranging (LiDAR) multi- and hyperspectral devices and drones, will also provide new capabilities in the field of mangrove species diversity monitoring (Fassnacht et al., 2016).

3.3 | Predicting species diversity

Predicting current and future mangrove species diversity is essential to faster processing and to detect shared patterns of environmental response across various communities and enhancing the capacity to synthesize complex data into a ready to use information for scientists and decision-makers (Ferrier & Guisan, 2006). Modelling approaches, such as stacked species distributions (Ferrier & Guisan, 2006), macroecological and ordination (Dubuis et al., 2011), generalized additive models (Sarker et al., 2019), as well as stochastic methods have been used to elucidate mangrove species spatial distribution patterns, stand distribution, richness, and compositions (Mateo et al., 2017; Sarker et al., 2019). A key factor for a successful model setup and validation is reliable spatially continuous information on the relevant environmental variables combined with species diversity and structure distribution in the stand. To this end, field inventories provide the most accurate information, but only yield a snapshot of data, as frequent observations are seldom possible. However, the spatial species distribution modelling is still limited to extrapolating small amounts of observed field data to larger spatial scale, validation, and uncertainties. The lack of adequate field inventories and data on spatiotemporal changes has constrained the effectiveness and application of modelling approaches (Berger et al., 2008; Rivera-Monroy et al., 2017). Satellite-based remote sensing provides approaches to quantify long-term monitor the structure, species diversity, and temporal dynamics (Lucas et al., 2017; Lucas et al., 2020; Otero et al., 2019; Pimple et al., 2022; Wang et al., 2019). The synergy of the individual- and process-based forest modeling approaches with remote sensing is a promising method (Fischer et al., 2019) although not adequately explored. In addition, the above-mentioned data issues and temporal inconsistencies (see sections 3.1 and 3.2) may limit the performance of such integrated approaches.

4 | PRACTICES FOR TREE SPECIES DIVERSITY MONITORING AND PREDICTIONS

4.1 | Remote sensing-based systematic sampling design for characterizing tree species diversity

The assemblage of mangrove trees species at any site is attributed to the different responses of local environmental settings such as variability in topography, tidal regimes, coastal processes, canal networks, and edaphic conditions. In addition to natural factors, anthropogenic modification such as rehabilitation practices, play an important role in modifying landscape scale species distribution (Das et al., 2019; Ellison et al., 2000). However, ecological baseline species distribution studies have rarely considered the effects of various factors at the landscape scale.

Three studies have previously described good sampling practices while reporting on species distribution in mangrove forests (Castaneda-Moya et al., 2006; Dale, 1999; Ellison et al., 2000) stating that: (1) sampling must be performed in contiguous quadrats or areas along a transects perpendicular to the shoreline, (2) a comprehensive list of species for the study site is required, (3) measurements of each species abundance (i.e., where they occur in the given landscape) are required, and (4) measurements of the edaphic parameters and surface elevation are also essential. Dale (1999) noted that in species distribution studies, researchers should describe and quantify spatiotemporal characteristics and then relate them to observed characteristics and processes. In addition to the four strategies mentioned above (as a minimum), good sampling strategies must also consider the local anthropogenic settings and temporal characteristics (Pimple et al., 2022). Figure 4 shows a conceptual outline of good sampling practices and required parameters when studying mangrove species distribution patterns based on the present review. To achieve this, a systematic sampling design across mangrove forest landscapes is crucial; however, field-based observations alone are impractical for large-scale sampling designs. Accordingly, advances in remote sensing create new possibilities for overcoming the limitations of traditional sampling practices.

In traditional mangrove ecological studies, remote sensing-based spatiotemporal dimensions have not been explored for distribution

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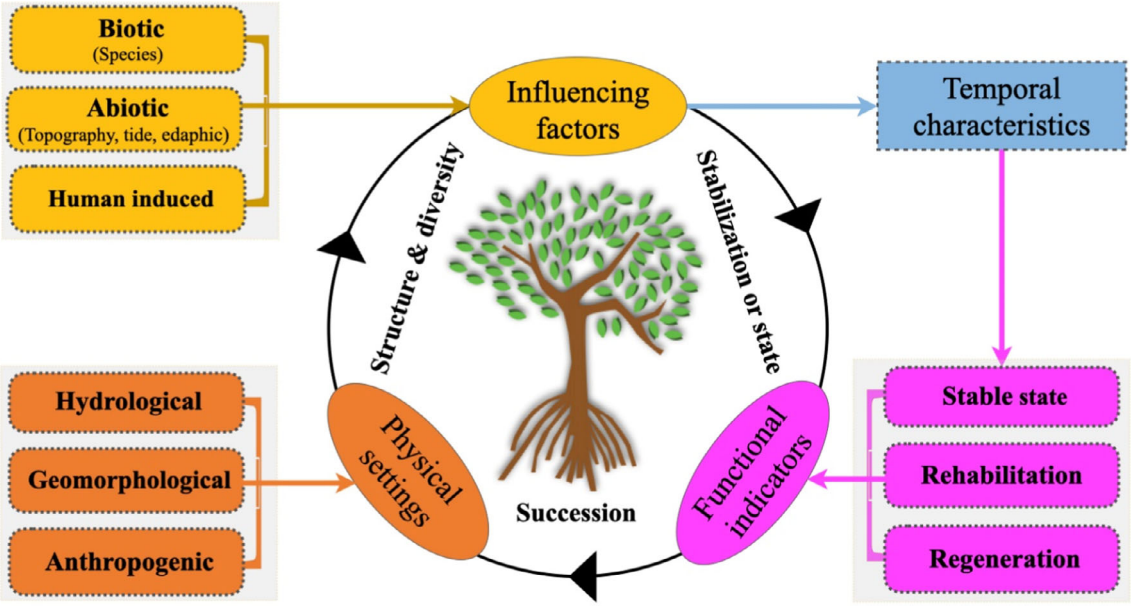

FIGURE 4 Good sampling strategy practices and required parameters for monitoring spatial scale mangrove tree species diversity distribution using contiguous quadrats or area sampling along a transect: Orange boxes indicate the physical factors related to environmental and anthropogenic settings, yellow boxes indicate the natural and anthropogenic influential factors, and pink boxes indicate the possible temporal patterns that could be observed over time [Colour figure can be viewed at wileyonlinelibrary.com]

analyses. Remote sensing-based spatially balanced sample approaches, such as area proportional stratified random sampling, have been previously investigated (Grafström et al., 2014; Köhl et al., 2006; Olofsson et al., 2014; Wallner et al., 2018). In such inventory designs, the study site can be partitioned into predefined clusters, where the proportion of forest species can be measured using spectral (i.e., multispectral) and structural (i.e., SAR) properties acquired via remote sensing imagery. Indeed, distinct spectral and structural heterogeneity of forest landscapes could be helpful prior to sampling for determining the study design, identifying possible locations of transect lines or sampling plots (Pimple et al., 2022). The spectral response fluctuation in LANDSAT and SENTINEL-2 time-series can reflect the various phenomena that occur at specific sampling locations (Otero et al., 2019; Pimple et al., 2020; Pimple et al., 2022). Furthermore, quantification or statistical testing of distribution (Clarke et al., 2008; Ellison et al., 2000) based on systematic sampling designs and inclusion of environmental factors can yield more reliable outcomes than traditional random sampling approaches (Pimple et al., 2022). Such innovative sampling designs enable a higher inventory precision, lower sample number, reliable locations to establish transects, and an understanding of temporal changes. There is considerable uncertainty in the upscaling of small or non-systematic species diversity field measurements to the larger spatial scales observed in remote sensing data and modelling studies; thus, systematically determined spectral and structural variability based on tree species compositions, as well as field samples, can be further used in remote sensing-based classification approaches, accuracy assessments, and parameterization of modelling studies. Low-cost or freely available high-resolution imagery (e.g., SENTINEL-SAR, SENTINEL-2 MSI, PLANETSCOPE, and drones) offers the potential to significantly improve the sampling design while reducing the associated costs.

4.2 | Overcoming remote sensing data and tree species identification challenges

4.2.1 | Obtaining consistent quality remote-sensing data

Most previous studies have used single date, cloud-free imagery to determine mangrove forest extent and spatial distribution; however, recent time series-based compositing approaches of medium-to-high resolution imagery via the availability of open-source data that is free-to-use for research and education, in addition to novel non-profit cloud-computing platforms, such as the Google Earth Engine (GEE), are facilitating large-scale environmental monitoring, analysis, pre-processing (such as algorithms to remove the cloud cover and shadows), and processing of high volumes of earth observation imagery (Gomes et al., 2020; Hauser et al., 2020; Simonetti et al., 2021). However, sparse canopy, low-height, or scrub mangroves are likely to be submerged or inundated during high tides (Li et al., 2019; Xia, Qin, et al., 2020); thus, the consistency and quality of derived data may vary depending on pixels observed during low or high tide.

Accordingly, the favored utilization of low tide imagery allows for fully or partially submerged mangroves to be distinguished. Additional studies have demonstrated approaches to extract low tide acquisitions from remote sensing data (Murray et al., 2012), such as obtaining a multi-resolution tidal modelling framework from LANDSAT imagery

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during low tide (Sagar et al., 2017; Sagar et al., 2018). Accordingly, image acquisition time can be filtered using local tidal gauge station data (if available) and further cross-checked using the normalized difference water index (NDWI; Murray et al., 2012). Furthermore, this approach could be extended to high-resolution time- series (e.g., SENTINEL-1(SAR)-2 (MSI), PLANETSCOPE, or commercial high-resolution imagery) to improve mangrove ecosystems monitoring.

Studies have indicated that the prior stratification of an area of interest (i.e., the separation of mangrove area) can reduce spectral confusion, especially if the subject is terrestrial forest, other forms of green vegetation, or cultivated fields (Long & Giri, 2011; Pimple et al., 2017). In addition, the Global Mangrove Watch dataset can also be used for stratification at national or regional scales (Bunting et al., 2022). Accordingly, the stratified mangrove area of interest can be directly used as a subset for high-resolution imagery (preferably at low tide acquisition times), to enhance the spectral (i.e., multispectral) and structural (i.e., SAR) properties.

4.2.2 | Trajectories of change

Spectral variability of time-series imagery can serve as a powerful proxy for temporal changes or species succession. Here, the focus has been placed on reviewing recent literature containing time-series analyses for deriving a set of possible conditions characterizing the time and magnitude of various natural and anthropogenic disturbances (Chazdon et al., 2016; Otero et al., 2019; Pimple et al., 2020; Pimple et al., 2022; Ryo et al., 2019; Wulder et al., 2019). Observed changes across time could be identified, studied, and used as a proxy for interpreting the observed diversity and structural differences as they relate to underlying stressors or processes. Furthermore, in complex, diverse landscapes (where many species co-exist), temporal assessments could be used to classify the landscape into natural, rehabilitated, or regenerated forest types. Based on the above literature, satellite-based vegetation indices [e.g., normalized difference infrared index (NDII)] can be used to characterize the temporal stages as they relate to mangrove species diversity (Figure 5).

1. *Undisturbed forest stands*: Undisturbed, mature mangrove forest stands have relatively stable values of spectral indices values over time. In this state, mangroves are expected to be more structurally complex and diverse.
2. *Rehabilitated forest stands*: Spectral indices values are expected to gradually increase and then return (or come close) to those of the undisturbed forest stands before the disturbance. Rehabilitated mangroves are typically monoculture stands and less structurally complex, a potentially important indicator when evaluating rehabilitation project success.
3. *Regenerated forest stands*: Spectral indices values are expected to increase gradually and return (or come close) to those of the undisturbed forest stands before the disturbance. In these forest stands, the disturbance is normally followed by natural recovery without any external rehabilitation or restoration efforts, thus indicating post-disturbance recovery. Regenerated stands can be more diverse and structurally complex. A careful assessment of such sites is required, as its final tree species status may vary depending on the local environmental settings and available parent trees.
4. *Permanently disturbed forest stands*: Spectral indices values drop abruptly, with a clear breakpoint, and no gradual increase or return is observed over time. Forest stands are subjected to permanent loss of species diversity.

Consistent annual composites (Figure 5) are dependent on the availability of good pixels throughout the year. In regions with no available data due to consistent cloud cover, multi-year composites can be generated according to user-predefined rules (White et al., 2014). Furthermore, the SENTINEL-1 SAR and SENTINEL-2 MSI-based dense time-series may be capable of extracting structural variations, along with spectral proxies over time. Many recently developed cloud-computing platforms, such as GEE, Amazon Web Services, the System for Earth Observation Data Access, Processing and Analysis for Land Monitoring, the Joint Research Center Earth Observation Data and Processing Platform, and Data and Information Access Service are hosting entire archives of various remote sensing data, offering high-performance data processing capabilities (Cárdenas et al., 2017; Corbane et al., 2020; Gomes et al., 2020). These computing platforms can overcome conventional limitations associated with data selection, downloading, raw data pre-processing, image processing, and classification applications over larger scales.

4.3 | Linking spatial scale tree species models with remote sensing

Recent models have increasingly focused on spatial gradients of the abiotic environment (e.g., fertility, elevation, hydrology) and how heterogeneity is reflected in spatiotemporal vegetation patterns (Rivera-Monroy et al., 2017). The new modeling frameworks focus on the prediction and development of heterogeneous natural systems. Examples of this trend include investigations of regime shifts and distribution patterns (Bathmann et al., 2021; Jiang et al., 2012; Peters et al., 2020). However, such modelling faces data challenges including sampling collection strategies, characterization of species distribution, structure, and variations in topography. Remote sensing data have the potential to be used in either parameterizing or evaluating mechanistic mangrove stand models. Developments in remote sensing and drone-based methods are providing an increased potential for measurements of forest structure, species composition, biomass and productivity patterns over time and space, and the data on the assessment of performance measurements in mangrove rehabilitation projects to improve the prediction of mangrove forest diversity and dynamics at a large spatiotemporal scale (Pimple et al., 2020; Pimple et al., 2022; Shugart et al., 2015; Shugart et al., 2018). Several studies have described remote sensing applications such as SAR and multispectral data for mangrove structure and diversity monitoring (Fatoyinbo et al., 2008; Rahman et al., 2019; Simard et al., 2006; Wang et al., 2019). Peters et al. (2020) indicated another benefit of coupling modelling with remote sensing approaches. Furthermore, Pimple et al.

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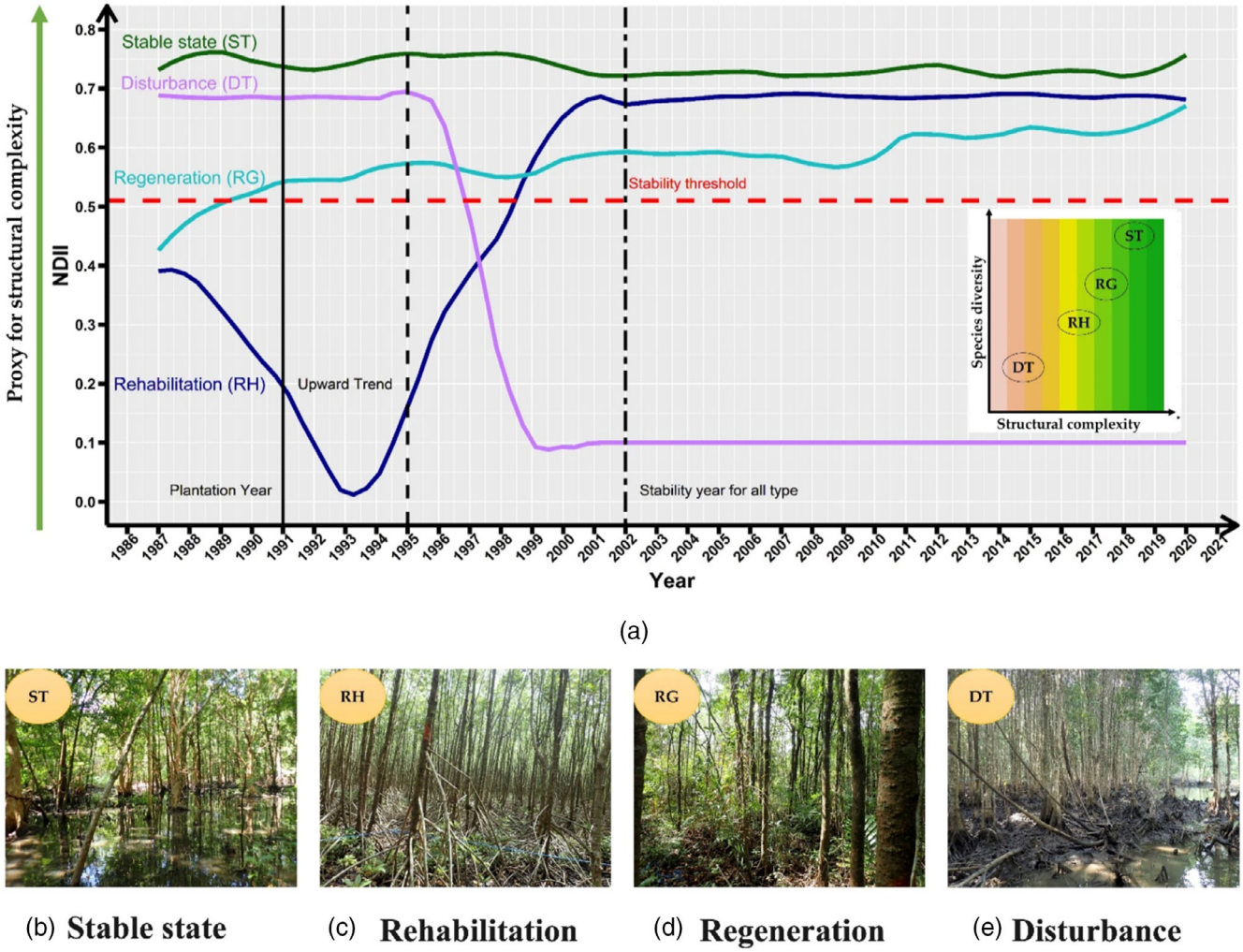

FIGURE 5 Normalized difference infrared index (NDII)-based temporal characteristics of forest stands, as derived from actual field data collected in Trat Province, Thailand: ST, stable or undisturbed state (dark green); RH, rehabilitation (dark blue); RG, regeneration (light blue); and, DT, disturbance with no recovery (purple). Known plantation year is indicated by a solid black vertical line. The start of the NDII upward trend following rehabilitation is indicated by a first black vertical dashed line (upward trend), whereas the quasi-stability years for all forest types are indicated by two vertical dashed lines (from upward trend to stability threshold). In the inset graph of structural complexity and species diversity, the brown colours indicate lower diversity and structural complexity, while the greener colors indicate higher diversity and structural complexity (left to right). Subpanels b, c, d, and e present the field plot photos of stable mature, rehabilitated, regenerated, and disturbed forest stands [Colour figure can be viewed at wileyonlinelibrary.com]

(2020) noted the possibility of using LANDSAT and SENTINEL-2 MSI time-series data to validate the performance of landscape and individual-based models for rehabilitation projects. Despite the advantages of remote sensing technology, uncertainties remain in parameterizing or evaluating the mechanistic models and directly influence the vegetation properties if above-mentioned data issues are not addressed. These spatial scale models could benefit greatly from the enhanced information gained from remote sensing data.

5 | SUMMARY OF PRACTICES

The literature review on mangrove tree species diversity monitoring reports that most studies: (1) are based on field inventories and have focused on small sites, (2) lacked systematic field sampling components, (3) have not attempted to quantify or statistically relate distribution patterns, and (4) neglected temporal dynamics when assessing diversity. Fassnacht et al. (2016) discussed several issues related to field data collection in forestry, indicating that the sampling strategy and species class criteria play crucial roles when reporting forest diversity. Indeed, remote sensing-based stratified sampling yields more precise diversity estimates when spectrally and structurally homogeneous and individual strata (classes) are identified (Wallner et al., 2018). To this end, the consistency and quality of remote sensing data can be improved by the consistent use of low tide imagery (Rogers et al., 2017; Sagar et al., 2017). The integration of remote sensing data can also be used to improve the parameterization or evaluation of large-scale modelling studies. An outline of best practices based on the present review is shown in Figure 6.

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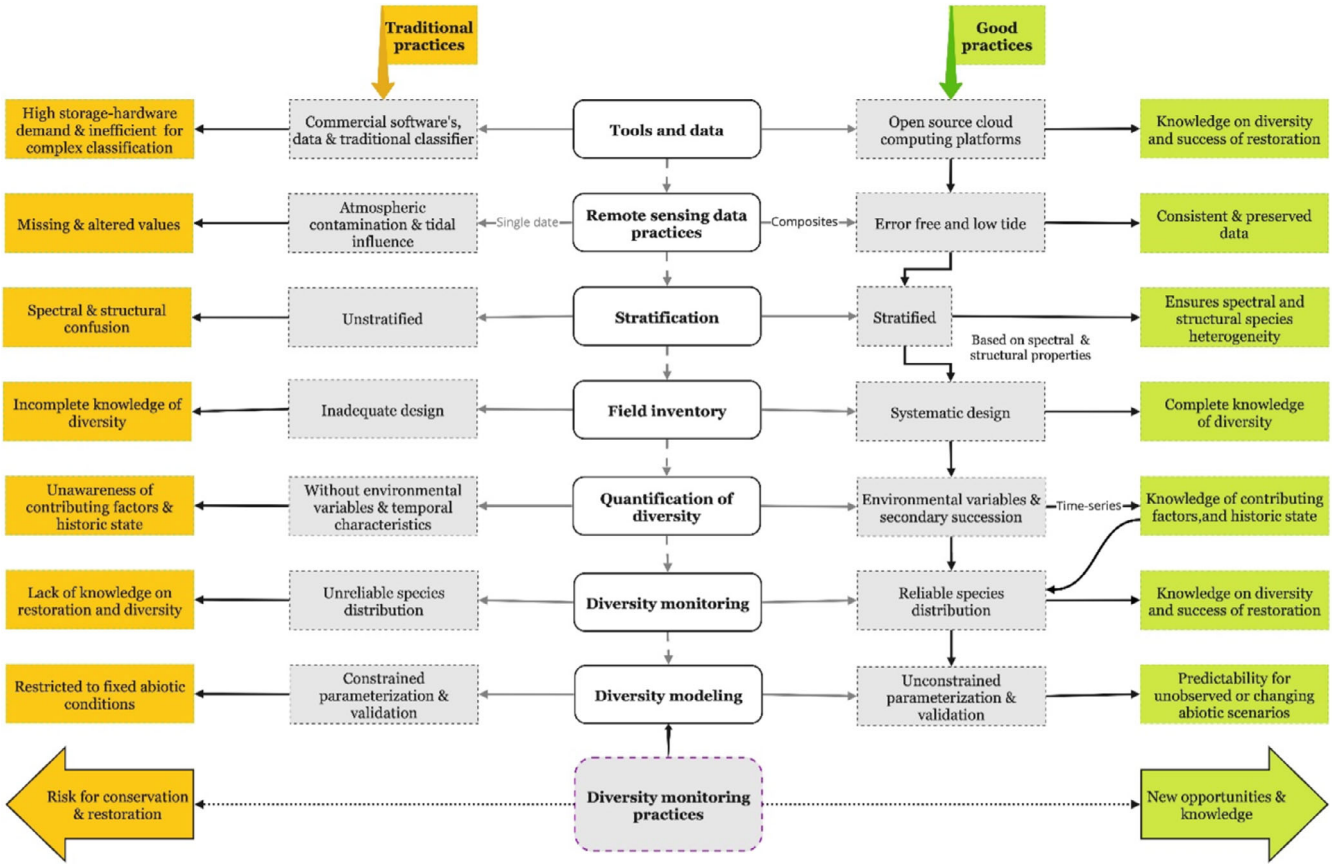

FIGURE 6 Conceptual diagram highlighting the comparison between reviewed traditional and proposed good practices using earth observation data. The general methodological steps for diversity monitoring are indicated by white rectangle boxes. The specific methods, data, and tools are indicated by gray boxes. Orange and green boxes indicate the traditional and best practices, respectively. The orange and green arrow indicates the outcome of traditional and proposed best practices, respectively [Colour figure can be viewed at wileyonlinelibrary.com]

Practice 1. Selection of low tide image: The selection of time-series based on low tide imagery allows the identification of submerged mangroves and maintains spectral and structural heterogeneity across different forest species. In multi-year analyses, radiometric normalization (dark object subtraction) can be performed to reduce variations caused by atmospheric conditions during image acquisition.

Practice 2. Stratification of mangrove and non-mangrove area: The delineation of mangroves as an area of interest can reduce spectral and structural confusions caused by other vegetation categories. To this end, freely available Global Mangrove Watch dataset, DEMs and vegetation indices (e.g., NDVI, NDII, and NDWI) are useful for detecting low elevation vegetation, such as mangroves.

Practice 3. Systematic sampling design: A systematic sampling design must include stratified transect plots perpendicular to the coastline, as well as distributed plots across the landscape. Such a sampling design is crucial for characterizing the site-specific ecological distribution of all available species. Furthermore, remote sensing-based spatial distribution mapping could greatly benefit from this information. This integration of structural and spectral information from multi-source data, such as optical, SAR, and drones would be highly beneficial for establishing thematic categories efficiently select the locations, transect, and sampling plots across the landscape.

Practice 4. Quantification of diversity distribution: Accurate quantification or statistical testing of species distribution in relation to various environmental variables (e.g., topography, tidal zones, and edaphic factors) is essential. In addition, the inclusion of anthropogenic effects, such as alterations to elevation or hydrology in modified landscapes (e.g., shrimp farms, rehabilitated mangroves), is important. Such quantifications will enable the definition of mangrove tree species distributions, such as single species zones, grouped, or random placement.

Practice 5. Characterization of temporal dynamics: The temporal characteristics of tree species patterns may vary between sites according to land modifications, rehabilitation practices, and various natural stressors. To this end, local changes could be studied through time and used as a proxy to interpret recent observations of structure and diversity (Figure 5). Tree diversity of undisturbed forests is expectedly characterized by stable spectral values, with more diverse and structurally complex biodiversity. Comparatively, rehabilitated forests are expected to gradually grow and align more closely with undisturbed forests, whereas

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rehabilitated stands are usually monocultures with lower structural complexity. Typically, the spectral values of regenerated stands gradually return to those of the undisturbed forest stands. Therefore, assessing the temporal dynamics is key to determining the success of rehabilitation projects. However, it is noteworthy that diversity and structural characteristics may vary depending on local conditions.

6 | CONCLUSIONS

Regardless of the current technological advancements, standardized reporting and monitoring methods for the mangrove tree species habitat are lacking, as are reliable scientific data to inform mangrove conservation interventions. The current review has identified data issues focusing on site-specific field inventories, as well as remote sensing- and modelling-based mangrove species characterization. Field-inventory-based species diversity studies limit the information about the criteria to choose the location of transects and plots and capture study sites spatial species heterogeneity. In addition, spatially unbalanced training and validation data affect spatial scale remote sensing-based species identifications. Notably, remote sensing-based (optical and SAR) imagery of mangrove species identification is limited by various environmental (e.g., tidal inundation) and atmospheric contaminants (e.g., cloud cover) in coastal areas, which results in hindering the measurement data quality and yielding unreliable estimates. Further, the efficacy and application of modeling-based approaches have been constrained by inadequate field inventories and data on temporal changes.

To this end, remote sensing-based mangrove stratification is imperative to aid systematic sampling design. Optical and SAR-based methods could help define heterogeneous mangrove species distribution at larger spatial scales. Such stratification based on a systematic sampling design, in conjunction with quantification or statistical testing of spatial distribution patterns, could vastly improve the reliability of results for ecological and prediction studies. Further, cloud-free and low tide imagery can greatly enhance the spectral and structural discrimination performance of freely available medium and high-resolution satellite imagery. Besides, error free time series data could provide comprehensive information on the historical state and temporal dynamics of conservation efforts. In addition, modelling frameworks could benefit considerably from accurate information on the spatiotemporal evolution of mangrove diversity.

AUTHOR CONTRIBUTIONS

Uday Pimple: Conceptualization, methodology, writing original draft-review and editing, project administration, funding acquisition. **Dario Simonetti**: Methodology, formal analysis, review & editing. **Ronny Peters**: Formal analysis, resources, review and editing. **Uta Berger**: Project administration, funding acquisition, review and editing. **Erika Podest**: Editing, resources. **Valery Gond**: Project administration, funding acquisition, review.

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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